

Single core Copper & Aluminum Conductors

XLPE Insulated, Aluminum Tape armored and PVC Sheathed

Voltage Grade

3.6 / 6 (7.2) KV

Description

Soft annealed Stranded circular compacted Copper or Stranded circular compacted Aluminum conductor, semiconducting layer as conductor screen, XLPE insulated semiconducting layer as non metallic insulation screen, Copper tape or wire as metallic insulation screen, armored with aluminum tape and PVC sheathed.

Application

These cables are generally suitable for direct burial or for installation on trays or in ducts.

Technical Data

Relevant Standard	IEC 60502
Conductor	Plain copper conductor class 2 or Aluminum conductor class 2
Conductor Screen	Extruded Semiconductive Compound
Insulation	Extruded XLPE Compound Rated 90° C
Insulation Screen	Non Metallic part: Extruded Semiconductive Compound (Bonded or Strippable) to the insulation Metallic part: (by one of the following) • Copper tape is applied helically with a suitable overlap. • Copper Wires helically applied and banded with a Copper tape to achieve electrical contact • Overlapped Copper Tape then Copper Wires are also available (in single core only) Conductor screen, XLPE insulation and non metallic insulation screen are applied at the same time using triple head extruder.
Inner Sheath	Extruded PVC Compound
Armoring	Double Aluminum Tapes applied helically over the inner sheath.
Outer Sheath	Extruded PVC Compound
Outer Sheath Color	Black (default color), or any other color according to customer request
Maximum Operating Temp.	90° C
Minimum bending Radius	12 x outer diameter
Packing Condition	Our standard length on Wooden Drum / or according to customer request



Product Code	Nominal Cross Sectional Area	Max. Conductor Resistance			Operating Capacitance	Inductance		Current Rating				Approx. Overall Diameter	Approx. Weight
		DC at 20 °C	AC at 90° C			Flat	Trefoil	Laid in Ground		Laid in Free Air			
			Flat	Trefoil				Flat	Trefoil	Flat Touched	Trefoil		
	mm²	Ω/km	Ω/km	Ω/km	μF/km	mH/km	mH/km	A	A	A	A	mm	kg/km
A - Copper Conductors													
PM-CT14-X5-01-DP	50	0.3870	0.4937	0.4937	0.3425	0.4633	0.4171	224	218	241	236	24.4	1065
PM-CT15-X5-01-DP	70	0.2680	0.3420	0.3421	0.3975	0.4380	0.3917	276	267	302	295	26.2	1300
PM-CT16-X5-01-DP	95	0.1930	0.2466	0.2467	0.4432	0.4210	0.3748	328	316	366	358	27.9	1580
PM-CT17-X5-01-DP	120	0.1530	0.1957	0.1959	0.4858	0.4048	0.3586	372	361	420	412	29.3	1830
PM-CT18-X5-01-DP	150	0.1240	0.1589	0.1591	0.5314	0.3925	0.3463	410	402	474	469	30.8	2200
PM-CT19-X5-01-DP	185	0.0991	0.1275	0.1277	0.5860	0.3800	0.3338	461	454	545	540	32.6	2575
PM-CT20-X5-01-DP	240	0.0754	0.0976	0.0979	0.6311	0.3676	0.3214	530	528	642	636	35.3	3170
PM-CT21-X5-01-DP	300	0.0601	0.0785	0.0790	0.6466	0.3602	0.3140	588	588	733	728	38.1	3825
PM-CT22-X5-01-DP	400	0.0470	0.0623	0.0630	0.6722	0.3530	0.3068	655	662	835	842	41.6	4820
PM-CT23-X5-01-DP	500	0.0366	0.0498	0.0507	0.7084	0.3444	0.2981	720	737	964	969	45.5	5960
PM-CT24-X5-01-DP	630	0.0283	0.0401	0.0412	0.8036	0.3369	0.2907	804	825	1073	1086	49.8	7340
PM-CT25-X5-01-DP	800	0.0221	0.0331	0.0345	0.9056	0.3281	0.2818	879	908	1210	1225	54.6	9235
B - Aluminum Conductors													
PM-AT14-X5-01-DP	50	0.6410	0.8219	0.8220	0.3548	0.4608	0.4145	171	170	185	184	24.8	795
PM-AT15-X5-01-DP	70	0.4430	0.5682	0.5682	0.4006	0.4394	0.3932	210	208	232	232	26.3	895
PM-AT16-X5-01-DP	95	0.3200	0.4106	0.4106	0.4493	0.4225	0.3762	251	248	281	278	28.1	1025
PM-AT17-X5-01-DP	120	0.2530	0.3248	0.3249	0.4889	0.4073	0.3611	285	282	324	324	29.4	1130
PM-AT18-X5-01-DP	150	0.2060	0.2646	0.2647	0.5405	0.3935	0.3472	317	315	366	366	31.1	1330
PM-AT19-X5-01-DP	185	0.1640	0.2110	0.2111	0.5951	0.3811	0.3349	357	358	422	426	32.9	1480
PM-AT20-X5-01-DP	240	0.1250	0.1612	0.1614	0.6311	0.3676	0.3214	417	414	506	498	35.3	1715
PM-AT21-X5-01-DP	300	0.1000	0.1293	0.1296	0.6466	0.3602	0.3140	459	463	581	574	38.1	1980
PM-AT22-X5-01-DP	400	0.0778	0.1012	0.1016	0.6722	0.3530	0.3068	517	527	672	670	41.6	2455
PM-AT23-X5-01-DP	500	0.0605	0.0796	0.0802	0.7084	0.3444	0.2981	595	602	782	781	45.5	2910
PL-AT24-X5-01-DP	630	0.0469	0.0627	0.0635	0.8059	0.3367	0.2905	670	679	894	895	49.9	3475
PL-AT25-X5-01-DP	800	0.0367	0.0505	0.0515	0.9056	0.3281	0.2818	749	761	1025	1025	54.6	4170

The above data is approximate and subject to manufacturing tolerances

Note:-

Flame retardant PVC sheathed or Polyethylene Sheathed are available according to customer request
Longitudinal or radial Water tightness layers are also available according to customer request
the below data are applicable for cables with rated voltage 3.8 / 6.6 (7.2) KV